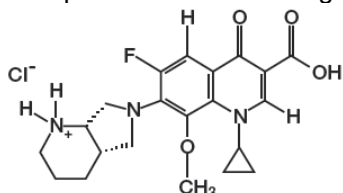


VIGAMOX®

(moxifloxacin hydrochloride ophthalmic solution) 0.5% as base

DESCRIPTION: VIGAMOX (moxifloxacin HCl ophthalmic solution) 0.5% is a sterile topical ophthalmic solution. Moxifloxacin is a fourth-generation fluoroquinolone antibacterial agent active against a broad spectrum of Gram-positive and Gram-negative ocular pathogens, atypical microorganisms and anaerobes.



C₂₁H₂₄FN₃O₄•HCl Mol Wt 437.9

Chemical Name:

1-Cyclopropyl-6-fluoro-1,4-dihydro-8-methoxy-7-[(4aS,7aS)-octahydro-6H-pyrrolol[3,4-b]pyridin-6-yl]-4-oxo-3-quinolinecarboxylic acid, monohydrochloride. Moxifloxacin differs from other quinolones in that it has a methoxy function at the 8 position, and an S,S- configured diazabicyclononyl ring moiety at the 7- position. Moxifloxacin hydrochloride is a slightly yellow to yellow crystalline powder.

Each mL of VIGAMOX Solution contains 5.45 mg moxifloxacin hydrochloride equivalent to 5 mg moxifloxacin base.

Contains: Active: Moxifloxacin 0.5% (5 mg/mL); **Preservative:** None. Product is self-preserved.

Inactives: sodium chloride, boric acid and purified water. May also contain hydrochloric acid/sodium hydroxide to adjust pH. VIGAMOX Solution is isotonic and formulated at pH 6.8 with an osmolality of approximately 290 mOsm/kg.

CLINICAL PHARMACOLOGY:

Pharmacokinetics/Pharmacodynamics: Following topical ocular administration of VIGAMOX Solution, moxifloxacin was absorbed into the systemic circulation. Plasma concentrations of moxifloxacin were measured in 21 male and female subjects who received bilateral topical ocular doses of VIGAMOX Solution 3 times a day for 4 days. The mean steady-state C_{max} and AUC were 2.7 ng/mL and 41.9 ng·hr/mL, respectively. These exposure values are approximately 1,600 and 1,200 times lower than the mean C_{max} and AUC reported after well-tolerated therapeutic 400 mg oral doses of moxifloxacin. The plasma half-life of moxifloxacin was estimated to be 13 hours.

Microbiology: Moxifloxacin has in vitro activity against a wide range of Gram-positive and Gram-negative microorganisms. Moxifloxacin inhibits the topoisomerase II (DNA gyrase) and topoisomerase IV required for bacterial DNA replication, transcription, repair, and recombination. The C8-methoxy moiety of moxifloxacin also lessens the selection of resistant mutants of Gram-positive bacteria compared to the C8-H moiety found in older fluoroquinolones. Moxifloxacin's bulky C-7 substituent group interferes with the quinolone efflux pump mechanism of bacteria. Moxifloxacin is often bactericidal at concentrations equal to or slightly greater than inhibitory concentrations.

Fluoroquinolones, including moxifloxacin, differ in chemical structure and mode of action from β-lactam antibiotics, macrolides and aminoglycosides, and therefore may be active against bacteria resistant to β-lactam antibiotics, macrolides and aminoglycosides. Therefore, organisms resistant to these drugs may be susceptible to moxifloxacin. In vitro resistance to moxifloxacin develops slowly via multiple-step mutations and occurs at a general frequency between 10⁻⁹ to 10⁻¹¹ for Gram-positive bacteria.

Moxifloxacin has been shown to be active against most strains of the following microorganisms, both in vitro and in clinical infections as described in the INDICATIONS AND USAGE section:

Gram-positive bacteria:

Corynebacterium species

Microbacterium species

Micrococcus luteus [including erythromycin, gentamicin, tetracycline, and/or trimethoprim resistant strains]

Staphylococcus aureus [including methicillin, erythromycin, gentamicin, ofloxacin, tetracycline and/or trimethoprim resistant strains]

Staphylococcus epidermidis [including methicillin, erythromycin, gentamicin, ofloxacin, tetracycline and/or trimethoprim resistant strains]

Staphylococcus haemolyticus [including methicillin, erythromycin, gentamicin, ofloxacin, tetracycline and/or trimethoprim resistant strains]
Staphylococcus hominis [including methicillin, erythromycin, gentamicin, ofloxacin, tetracycline and/or trimethoprim resistant strains]
Staphylococcus warneri [including erythromycin resistant strains]
Streptococcus mitis [including penicillin, erythromycin, tetracycline and/or trimethoprim resistant strains]
Streptococcus pneumoniae [including penicillin, erythromycin, gentamicin, tetracycline and/or trimethoprim resistant strains]
Streptococcus viridans [including penicillin, erythromycin, tetracycline and/or trimethoprim resistant strains]

Gram-negative bacteria:

Acinetobacter species
Haemophilus "alconae" [including ampicillin resistant strains]
Haemophilus influenzae [including ampicillin resistant strains]
Klebsiella pneumoniae
Moraxella catarrhalis
Pseudomonas aeruginosa

Other microorganisms:

Chlamydia trachomatis
Moxifloxacin has been shown to be active in vitro against most strains of the following organisms; however, the clinical significance of these data is unknown.

Gram-positive bacteria:

Arthrobacter species
Bacillus cereus
Bacillus thuringensis
Corynebacterium accolens
Corynebacterium amycolatum
Corynebacterium bovis
Corynebacterium macginleyi
Corynebacterium propinquum
Corynebacterium pseudodiphtheriticum
Enterococcus faecalis
Exiguobacterium species
Kocuria kristinae
Kocuria "lindaee"
Kocuria rhizophila
Listeria monocytogenes
Microbacterium "harmaniae"
Microbacterium "otitidis"
Rothia mucilaginosa
Staphylococcus arlettae
Staphylococcus capitis
Staphylococcus caprae
Staphylococcus cohnii
Staphylococcus lugdunensis
Staphylococcus pasteurii
Staphylococcus saprophyticus
Staphylococcus sciuri
Streptococcus agalactiae
Streptococcus "conjunctivae"
Streptococcus cristatus
Streptococcus dysgalactiae
Streptococcus mitis

Gram-negative bacteria:

Achromobacter xylosoxidans
Acinetobacter baumannii
Acinetobacter calcoaceticus
Acinetobacter johnsonii
Acinetobacter junii
Acinetobacter lwoffii
Acinetobacter "mumbaiae"
Acinetobacter schindleri
Acinetobacter ursingii
Aeromonas caviae
Chryseobacterium indologenes
Chryseobacterium species
Citrobacter freundii
Citrobacter koseri
Enterobacter aerogenes
Enterobacter cloacae
Enterobacter hormaechei
Escherichia coli
Klebsiella oxytoca
Moraxella osloensis
Morganella morganii
Neisseria gonorrhoeae
Pantoea agglomerans
Proteus mirabilis
Proteus vulgaris
Pseudomonas oryziatrans
Pseudomonas stutzeri
Serratia liquefaciens *Serratia marcescens*
Stenotrophomonas maltophilia

Anaerobic microorganisms:

Clostridium perfringens
Fusobacterium species
Porphyromonas species
Prevotella species
Propionibacterium acnes

Other Organisms:

Atypical Mycobacterium
Chlamydia pneumoniae
Legionella pneumophila
Mycobacterium avium
Mycobacterium marinum
Mycoplasma pneumoniae

Streptococcus Groups C, G and F
Streptococcus "ocularis"
Streptococcus oralis
Streptococcus parasanguinis
Streptococcus pyogenes
Streptococcus salivarius
Streptococcus sanguis
Streptococcus "schlechii"

Clinical Studies: VIGAMOX Solution has been studied in patients from newborns to adults, including geriatric patients.

In three randomized, double masked, multicenter, controlled clinical trials in which patients were dosed 3 times a day for 4 days, VIGAMOX Solution produced clinical cures in 80% to 94% of patients treated for bacterial conjunctivitis. Microbiological success rates for the eradication of the baseline pathogens ranged from 85% to 97%.

In one of these trials in pediatric patients from birth to one month of age, VIGAMOX Solution produced clinical cure in 80% of patients with bacterial conjunctivitis. The microbiological success rate for the eradication of the baseline pathogens was 92%.

In a randomized, double-masked, multicenter, controlled clinical trial in which patients were dosed 2 times a day for 3 days, VIGAMOX Solution produced clinical cure in 74% of patients treated for bacterial conjunctivitis. Microbiological success rate for the eradication of the baseline pathogens was 81%.

Special Populations: The pharmacokinetic parameters of oral moxifloxacin are not significantly altered by mild, moderate or severe renal impairment. No dosage adjustment of VIGAMOX Solution is necessary in patients with renal impairment.

Pharmacokinetic parameters of oral moxifloxacin were not significantly altered in patients with mild to moderate hepatic insufficiency (Child Pugh Classes A and B). Studies were not performed in patients with severe hepatic impairment (Child Pugh Class C). Because of the low systemic exposure by the topical route of administration, no dosage adjustment of VIGAMOX Solution is needed in patients with hepatic impairment.

INDICATIONS AND USAGE: VIGAMOX Solution is indicated for the treatment of patients 1 year of age and older with bacterial conjunctivitis caused by susceptible strains of the following organisms.

Gram-positive bacteria:

Corynebacterium species†

Microbacterium species

Micrococcus luteus† [including erythromycin, gentamicin, tetracycline, and/or trimethoprim resistant strains]

Staphylococcus aureus [including methicillin, erythromycin, gentamicin, ofloxacin, tetracycline and/or trimethoprim resistant strains]

Staphylococcus epidermidis [including methicillin, erythromycin, gentamicin, ofloxacin, tetracycline and/or trimethoprim resistant strains]

Staphylococcus haemolyticus [including methicillin, erythromycin, gentamicin, ofloxacin, tetracycline and/or trimethoprim resistant strains]

Staphylococcus hominis† [including methicillin, erythromycin, gentamicin, ofloxacin, tetracycline and/or trimethoprim resistant strains] *Staphylococcus warneri*† [including erythromycin resistant strains]

Streptococcus mitis† [including penicillin, erythromycin, tetracycline and/or trimethoprim resistant strains]

Streptococcus pneumoniae [including penicillin, erythromycin, gentamicin, tetracycline and/or trimethoprim resistant strains]

Streptococcus viridans [including penicillin, erythromycin, tetracycline and/or trimethoprim resistant strains]

Gram-negative bacteria:

Acinetobacter species

Haemophilus "alconae" [including ampicillin resistant strains]

Haemophilus influenzae [including ampicillin resistant strains]

Klebsiella pneumoniae†

Moraxella catarrhalis†

*Pseudomonas aeruginosa**

Other microorganisms:

Chlamydia trachomatis

†Efficacy for this organism was studied in fewer than 10 infections

Preoperative and postoperative sterilization (when prophylactic antibiotic treatment is required).

CONTRAINDICATIONS: VIGAMOX Solution is contraindicated in patients with a history of hypersensitivity to moxifloxacin, to other quinolones, or to any of the components in this medication.

WARNINGS: For ocular use only. Not for injection. VIGAMOX Solution should not be injected subconjunctivally or introduced directly into the anterior chamber of the eye. In patients receiving systemically administered quinolones, serious and occasionally fatal hypersensitivity (anaphylactic) reactions have been reported, some following the first dose. Some reactions were accompanied by cardiovascular collapse, loss of consciousness, angioedema (including laryngeal, pharyngeal or facial edema), airway obstruction, dyspnea, urticaria, and itching. If an allergic reaction to moxifloxacin occurs, discontinue use of the drug. Serious acute hypersensitivity reactions may require immediate emergency treatment. Oxygen and airway management should be administered as clinically indicated.

PRECAUTIONS:

General: As with other anti-infectives, prolonged use may result in overgrowth of non-susceptible organisms, including fungi. If superinfection occurs, discontinue use and institute alternative therapy. Whenever clinical judgment dictates, the patient should be examined with the aid of magnification, such as slit-lamp biomicroscopy, and, where appropriate, fluorescein staining. Patients should be advised not to wear contact lenses if they have signs and symptoms of bacterial conjunctivitis. Tendon inflammation and rupture may occur with systemic fluoroquinolone therapy including moxifloxacin, particularly in elderly patients and in those treated concurrently with corticosteroids. Therefore, treatment with VIGAMOX Solution should be discontinued at the first sign of tendon inflammation.

Information for Patients: Avoid contaminating the applicator tip with material from the eye, fingers or other source. Systemically administered quinolones have been associated with hypersensitivity reactions, even following a single dose. Discontinue use immediately and contact your physician at the first sign of a rash or allergic reaction.

Effects on ability to drive and use machines: Temporary blurred vision or other visual disturbances may affect the ability to drive or use machines. If blurred vision occurs at application, the patient must wait until the vision clears before driving or using machinery.

Drug Interactions: Given the low systemic concentration of moxifloxacin following topical ocular administration of the medicinal product, drug interactions are unlikely to occur. While drug-drug interaction studies have not been conducted with VIGAMOX Solution, they have been performed with the oral product at much higher systemic exposures than are achieved by the topical ocular route.

Unlike some other fluoroquinolones, no clinically significant drug-drug interactions between systemically administered moxifloxacin and itraconazole, theophylline, warfarin, digoxin, oral contraceptives, probenecid, ranitidine or glyburide have been observed. In vitro studies indicate that moxifloxacin does not inhibit CYP3A4, CYP2D6, CYP2C9, CYP2C19 or CYP1A2 indicating that moxifloxacin is unlikely to alter the pharmacokinetics of drugs metabolized by these cytochrome P450 isozymes.

Carcinogenesis, Mutagenesis, Impairment of Fertility: Moxifloxacin was not mutagenic in four bacterial strains used in the Ames Salmonella reversion assay. As with other quinolones, the positive response observed with moxifloxacin in strain TA 102 using the same assay may be due to the inhibition of DNA gyrase. Moxifloxacin was not mutagenic in the CHO/HGPRT mammalian cell gene mutation assay. An equivocal result was obtained in the same assay when v79 cells were used. Moxifloxacin was clastogenic in the v79 chromosome aberration assay, but it did not induce unscheduled DNA synthesis in cultured rat hepatocytes. There was no evidence of genotoxicity in vivo in a micronucleus test or a dominant lethal test in mice.

Moxifloxacin had no effect on fertility in male and female rats at oral doses as high as 500 mg/kg/day, approximately 21,700 times the highest recommended total daily human ophthalmic dose.

Long term studies in animals to determine the carcinogenic potential of moxifloxacin have not been performed. However, in an accelerated study with initiators and promoters, moxifloxacin was not carcinogenic following up to 38 weeks of oral dosing at 500 mg/kg/day.

Fertility: Studies have not been performed to evaluate the effect of ocular administration of VIGAMOX Solution on fertility.

Pregnancy: Teratogenic Effects.

Pregnancy Category C: Moxifloxacin was not teratogenic when administered to pregnant rats during organogenesis at oral doses as high as 500 mg/kg/day (approximately 21,700 times the highest recommended total daily human ophthalmic dose); however, decreased fetal body weights and slightly delayed fetal skeletal development were observed. There was no evidence of teratogenicity when pregnant Cynomolgus monkeys

were given oral doses as high as 100 mg/kg/day (approximately 4,300 times the highest recommended total daily human ophthalmic dose). An increased incidence of smaller fetuses was observed at 100 mg/kg/day. There are no or limited amount of data from the use of VIGAMOX Solution in pregnant women. However, no effects on pregnancy are anticipated since the systemic exposure to moxifloxacin from topical ocular application is negligible.

Since there are no adequate and well-controlled studies in pregnant women VIGAMOX Solution should be used during pregnancy only if the potential benefit justifies the potential risk to the fetus.

Nursing Mothers: Moxifloxacin/metabolites have not been measured in human milk, although it can be presumed to be excreted in human milk. Animal studies have shown excretion of low levels in breast milk after oral administration of moxifloxacin. However, at therapeutic doses of VIGAMOX Solution no effects on the suckling child are anticipated. Caution should be exercised when VIGAMOX Solution is administered to a nursing mother.

Pediatric Use: Safety and effectiveness in pediatric patients below the age of 1 year have not been established. There is no evidence that the ophthalmic administration of VIGAMOX Solution has any effect on weight bearing joints, even though oral administration of some quinolones has been shown to cause arthropathy in immature animals.

Geriatric Use: No overall differences in safety and effectiveness have been observed between elderly and other adult patients.

UNDESIRABLE EFFECTS:

The following adverse reactions have been reported during clinical trials with VIGAMOX Solution and are classified according to the following convention: very common (1/10), common (1/100 to <1/10), uncommon (1/1,000 to <1/100), rare (1/10,000 to <1/1,000), very rare (<1/10,000). Within each frequency grouping, adverse reactions are presented in order of decreasing seriousness.

System Organ Classification	Adverse reactions [MedDRA Preferred Term (v. 15.1)]
Blood and lymphatic system disorders	<i>Rare:</i> haemoglobin decreased
Nervous system disorders	<i>Uncommon:</i> headache <i>Rare:</i> paresthesia
Eye disorders	<i>Common:</i> eye pain, eye irritation <i>Uncommon:</i> punctate keratitis, dry eye, conjunctival haemorrhage, ocular hyperaemia, eye pruritus, eyelid oedema, ocular discomfort (burning/stinging) <i>Rare:</i> corneal epithelium defect, corneal disorder, conjunctivitis, blepharitis, eye swelling, conjunctival oedema, vision blurred, visual acuity reduced, asthenopia, <i>erythema of eyelid</i>
Respiratory, thoracic and mediastinal disorders	<i>Rare:</i> nasal discomfort, pharyngolaryngeal pain, sensation of foreign body (throat)
Gastrointestinal disorders	<i>Uncommon:</i> dysgeusia <i>Rare:</i> vomiting
Hepatobiliary disorders	<i>Rare:</i> alanine aminotransferase increased, gamma- glutamyltransferase increased

Additional adverse reactions identified from post marketing surveillance include the following. Frequencies cannot be estimated from the available data. Within each System Organ Class adverse reactions are presented in order of decreasing seriousness.

System Organ Classification	Adverse reactions [MedDRA Preferred Term (v. 15.1)]
Immune system disorders	hypersensitivity
Nervous system disorders	dizziness
Eye disorders	ulcerative keratitis, keratitis, lacrimation increased, photophobia, eye discharge
Cardiac disorders	palpitations
Respiratory, thoracic and	dyspnoea

mediastinal disorders	
Gastrointestinal disorders	nausea
Skin and subcutaneous tissue disorders	erythema, pruritus, rash, urticaria

DOSAGE AND ADMINISTRATION: Instill one drop in the affected eye 3 times a day for 7 days.

FOR PREOPERATIVE AND POSTOPERATIVE STERILIZATION: Usually, instill one drop in the affected eye(s) 5 times per day before operation, and 3 times per day after operation.

HOW SUPPLIED: VIGAMOX (moxifloxacin ophthalmic solution) 0.5% is supplied as a sterile ophthalmic solution in a natural low density polyethylene bottle, with dispensing plug and a white polypropylene closure. Tamper evidence is provided by a closure with an extended skirt that locks to the bottle finish on application and breaks away from the closure on opening.

Available in 5ml pack size.

OVERDOSAGE: No information is available on overdosage in humans. Due to the characteristics of this preparation no toxic effects are to be expected with an ocular overdose of the product, or in the event of accidental ingestion of the contents of one bottle. If a topical overdose of moxifloxacin 0.5% Ophthalmic Solution occurs, the eye(s) may be flushed with tap water. No reports of overdose were received during the clinical studies of moxifloxacin 0.5%.

Storage: Do not store above 30°C. Discard 28 days after opening.

Rx Only

PRODUCT REGISTRATION HOLDER

Novartis Corporation (Malaysia) Sdn. Bhd.

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